Decentralised Storage on Blockchain: IPFS, Filecoin, and Arweave

IPFS, Filecoin, and Arweave are three of the most popular decentralised storage solutions. Let's explore how each one works, what makes them different, and when to use them.

Most of us rely on cloud storage services like Google Drive or AWS to store our data. While these platforms are convenient they are centralised, which means your data lives on servers owned and controlled by a single company. That creates risks like downtime, censorship, security breaches, and rising storage costs.

Decentralised storage offers a new way forward. By spreading data across a network of users instead of one central location, it provides better privacy, resilience, and long-term efficiency.

Let's start with some basics of decentralised storage and then explore three of the most popular decentralised storage solutions: IPFS, Filecoin, and Arweave.

> ***"In a world where data is everything, decentralised storage offers the freedom to own, control, and share your files without the limitations of centralised authorities."***

What is decentralised storage?

Decentralised storage is a system where data is stored across a distributed network of nodes instead of on servers owned by a single company. Each file is broken into pieces, encrypted, and spread across multiple locations, making it more secure and resistant to tampering or loss.

IPFS (InterPlanetary File System)

The InterPlanetary File System (IPFS) is a peer-to-peer distributed file system developed by Protocol Labs. Its mission is to replace the traditional,

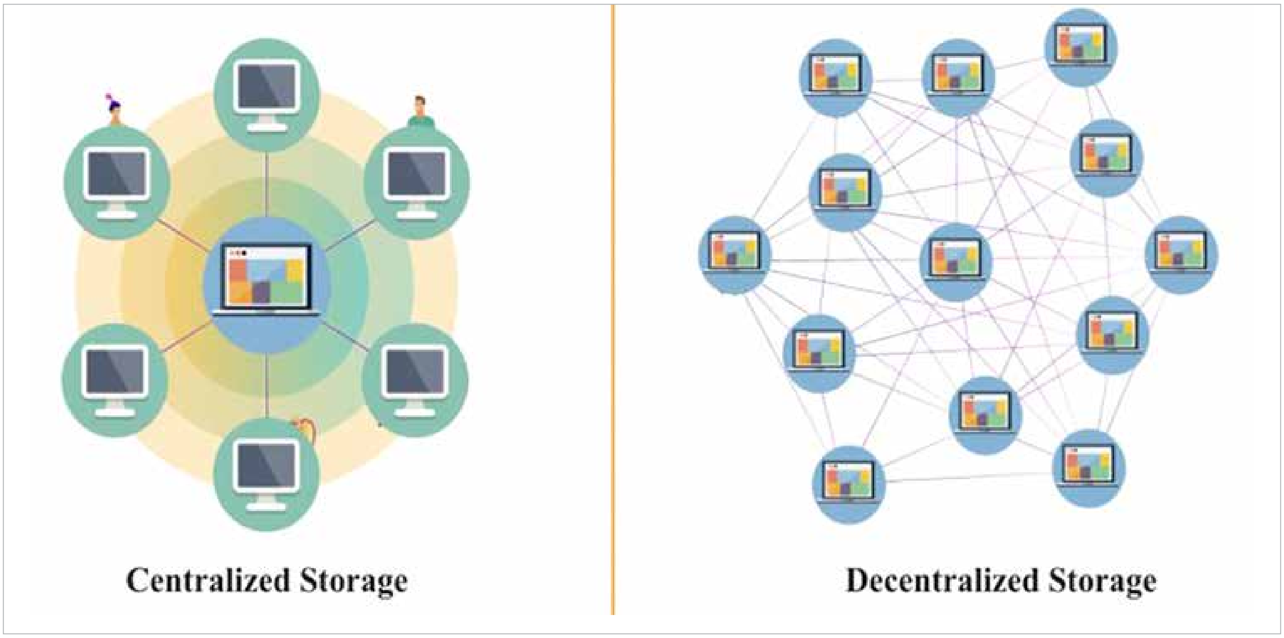

Figure 1: Storage